

GIAN-GABRIEL GARCIA

Curriculum Vitae

Department of Industrial & Systems Engineering
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EDUCATIONAL HISTORY

University of Michigan, Ann Arbor, MI
PhD, Industrial and Operations Engineering
July 2020

Dissertation: “Predictive and Prescriptive Analytics for Concussion Management Decisions”

University of Michigan, Ann Arbor, MI
MS, Industrial and Operations Engineering
December 2016

University of Pittsburgh, Pittsburgh, PA
BS, Industrial Engineering, *Summa Cum Laude*
December 2014

EMPLOYMENT HISTORY

University of Washington
Seattle, WA, USA
Assistant Professor, Industrial & Systems Engineering, 8/2025–Present

Georgia Institute of Technology
Atlanta, GA, USA
Harold E. Smalley Early Career Professor, Industrial and Systems Engineering, 8/2024-8/2025
Pillar 1 Co-lead, Pediatric Technology Center, 5/2024-8/2025
Assistant Professor, H. Milton Stewart School of Industrial and Systems Engineering, 8/2021-8/2025

Harvard Medical School
Boston, MA, USA
Postdoctoral Research Fellow, MGH Institute for Technology Assessment, 7/2020-7/2021

AWARDS AND HONORS

First Prize, INFORMS Minority Issues Forum Paper Competition, 2024
Outstanding Faculty Achievement in Research - Early Career, Georgia Tech ISyE, 2024
Annual CIOS Teaching Award, Georgia Tech, 2023
Diversity, Equity, and Inclusion Fellow, Georgia Tech ISyE, 2022
CIOS Honor Roll: Student Recognition of Excellence in Teaching, Georgia Tech, 2022-2024
Best Paper, IISE Transactions Focus Issue on *Operations Engineering & Analytics*, 2021

Towner Prize for Distinguished Academic Achievement, University of Michigan, 2020
First Prize, SMDM Lee B. Lusted Prize in *Quantitative Methods & Theoretical Developments*, 2020
First Prize, INFORMS Minority Issues Forum Paper Competition, 2020
First Prize, SMDM Lee B. Lusted Prize in *Quantitative Methods & Theoretical Developments*, 2018
INFORMS Bonder Scholarship for Applied OR in Health Services, 2018
First Prize, INFORMS Minority Issues Forum Poster Competition, 2017
NSF Graduate Research Fellowship, 2017-2020
Rackham Merit Fellowship, University of Michigan, 2015-2020

PUBLICATIONS

Refereed archival journal publications

- J1. C.-Y. Liao¹, **G.-G. P. Garcia**, K. Paynabar, Z. Dong, Y. Xie, and M. S. Jalali (2025). *Tides need STEMMED: A Locally Operating Spatio-Temporal Mutually Exciting point process with Dynamic network for Improving Opioid Overdose Death Prediction*. *Manufacturing & Service Operations Management* (Accepted)⁴
- J2. Y. Xie, **G.-G. P. Garcia**, E. Song, and N. Serban (2025). *Evaluating Access to Pediatric Psychosocial Services: A Discrete Event Simulation Approach under Uncertainty*. *Journal of Simulation* (Accepted)
- J3. L. L. Czerniak, **G.-G. P. Garcia**, M. W. Genthe, Y. Xia, M. S. Lavieri, M. A. McCrea, T. W. McAllister, P. F. Pasquina, S. W. Liebel, and S. P. Broglio (2025). *Prediction of symptom burden, cognitive status, and risk of psychological distress in NCAA athletes with sport-related concussion(s): findings from the NCAA-DoD CARE Consortium*. *Annals of Biomedical Engineering* (Accepted)
- J4. R. Parab, J. Feeley, M. Valero, L. Chadalawada, **G.-G. P. Garcia**, S. Sil Kar, A. Madabhushi, M. Breton, H. Shao, and F. J. Pasquel (2025). *How Artificial Intelligence and Machine Learning are Transforming Diabetes Care*. *Endocrine Practice* (Accepted)
- J5. Y. Yang¹, C.-Y. Liao¹, E. Keyvanshokoo, H. Shao, M. B. Weber, F. Pasquel, and **G.-G. P. Garcia** (2025). *A Responsible Framework for Assessing, Selecting, and Explaining Machine Learning Models in Cardiovascular Disease Outcomes Among People With Type 2 Diabetes: Methodology and Validation Study*. *JMIR Medical Informatics* **13**, e66200.
- J6. H. S. Pandey¹, B. Lahijanian², J. D. Schmidt, R. C. Lynall, S. P. Broglio, T. W. McAllister, M. A. McCrea, P. F. Pasquina, and **G.-G. P. Garcia** (2025). *Quantifying the Diagnostic Utility of Baseline Testing: An Analysis of the NCAA-DoD CARE Consortium Dataset*. *American Journal of Sports Medicine* **53**(1), 181–191
- J7. S. J. Lee¹, X. Gong³, and **G.-G. P. Garcia** (2024). *Modified Monotone Policy Iteration for Interpretable Policies in Markov Decision Processes and the Impact of State Ordering Rules*. *Annals of Operations Research* **347**, 783–841
- J8. S. Ma, A. Dehghanian, **G.-G. P. Garcia**, and N. Serban (2024). *Learning Hidden Markov Models with Structured Dynamics*. *INFORMS Journal on Computing* **37**(3), 531–556
- J9. **G.-G. P. Garcia**, L. N. Steimle, W. J. Marrero, and J. B. Sussman (2024). *Interpretable Policies and the Price of Interpretability in Hypertension Treatment Planning*. *Manufacturing & Service Operations Management* **26**(1), 80–94. ⁵

¹Graduate student advisee

²Postdoc advisee

³Undergraduate student advisee

⁴First Prize, 2022 SMDM Lee B. Lusted Prize in *Quantitative Methods & Theoretical Developments* awarded to Che-Yi Liao

⁵First Prize, 2024 INFORMS Minority Issues Forum Paper Competition

- J10. L. B. Lempke, A. J. Boltz, **G.-G. P. Garcia**, R. A. Syrydiuk, H. S. Pandey¹, M. A. McCrea, T. W. McAllister, and S. P. Broglio (2023). *Optimizing Baseline and Post-Concussion Assessments Through Factor Structure Analysis: Findings from the NCAA-DoD CARE Consortium*. *The Clinical Neuropsychologist* **38**(5), 1156–1174
- J11. **G.-G. P. Garcia**, L. L. Czerniak, M. S. Lavieri, S. W. Liebel, K. Van Pelt, P. F. Pasquina, T. W. McAllister, M. A. McCrea, S. P. Broglio, and CARE Consortium Investigators (2023). *Estimating the Relationship Between the Symptom-Free Waiting Period and Injury Rates after Return-to-Play from Concussion: A Simulation Analysis Using CARE Consortium Data*. *Sports Medicine* **53**(10), 2513–2528.
- J12. S. J. Lee¹, **G.-G. P. Garcia**, K. K. Stanhope, M. Platner, and S. L. Boulet (2023). *Interpretable machine learning to predict adverse perinatal outcomes: Examining marginal predictive value of risk factors throughout pregnancy*. *American Journal of Obstetrics and Gynecology MFM* **5**(10), 101096.
- J13. **G.-G. P. Garcia**, M. S. Lavieri, T. W. McAllister, M. A. McCrea, S. P. Broglio, and the CARE Consortium Investigators (2023). *Reducing the price of Naïveté in return-to-play from sports-related concussion*. en. *Production and Operations Management* **32**, 3081–3099. ⁶
- J14. S. W. Liebel, C. G. Turner, A. C. Svirsko, **G.-G. P. Garcia**, P. F. Pasquina, T. W. McAllister, M. A. McCrea, S. P. Broglio, and the CARE Consortium Investigators (2023). *Temporal Network Analysis of Neurocognitive Functioning and Psychological Symptoms in Collegiate Athletes after Concussion*. en. *Journal of Neurotrauma*, neu.2022.0431.
- J15. L. L. Czerniak, S. W. Liebel, H. Zhou, **G.-G. P. Garcia**, M. S. Lavieri, M. A. McCrea, T. W. McAllister, S. P. Broglio, and the CARE Consortium Investigators (2023). *Sensitivity and Specificity of the IMPACT Neurocognitive Test in Collegiate Athletes and United States Military Cadets with ADHD and/or LD: Findings from the NCAA-DoD CARE Consortium*. *Sports Medicine* **53**, 747–759.
- J16. K. E. Nolan, J. B. Caccese, A. P. Kontos, T. A. Buckley, **G.-G. P. Garcia**, N. Port, S. P. Broglio, T. W. McAllister, M. McCrea, P. F. Pasquina, J. P. Hayes, and on behalf of the CARE Consortium Investigators (2023). *Primary and Secondary Risk Factors Associated With Concussion Symptom Clusters in Collegiate Athletes: Results From the NCAA-DoD Grand Alliance CARE Consortium*. en. *Orthopaedic Journal of Sports Medicine* **11**(4), 232596712311635.
- J17. J. B. Caccese, **G.-G. P. Garcia**, A. P. Kontos, N. Port, K. Valerio, S. P. Broglio, M. McCrea, T. W. McAllister, P. F. Pasquina, T. A. Buckley, and CARE Consortium Investigators (2023). *Test–Retest Reliability and Efficacy of Individual Symptoms in Concussion Management*. en. *Clinical Journal of Sport Medicine* **33**(1), 52–60.
- J18. **G.-G. P. Garcia**, R. Dehghanpoor, E. J. Stringfellow, M. Gupta, J. Rochelle, E. Mason, T. A. Pujol, and M. S. Jalali (2022). *Identifying and Characterizing Medical Advice-Seekers on a Social Media Forum for Buprenorphine Use*. *International Journal of Environmental Research and Public Health* **19**(10), 6281.
- J19. **G.-G. P. Garcia**, E. J. Stringfellow, C. DiGennaro, N. Poellinger, J. Wood, S. Wakeman, and M. S. Jalali (2022). *Opioid overdose decedent characteristics during COVID-19*. *Annals of Medicine* **54**(1), 1081–1088.
- J20. C.-Y. Liao¹, **G.-G. P. Garcia**, C. DiGennaro, and M. S. Jalali (2022). *Racial Disparities in Opioid Overdose Deaths in Massachusetts*. *JAMA Network Open* **5**(4), e229081.
- J21. **G.-G. P. Garcia**, C. M. Schumb, M. S. Lavieri, H. Koffijberg, T. W. McAllister, M. A. McCrea, and S. P. Broglio (2022). *Developing Insights for Possible and Probable Acute Concussions Using Cluster Analysis*. *Journal of Neurotrauma* **39**(1-2), 102–113.
- J22. C. DiGennaro, **G.-G. P. Garcia**, E. J. Stringfellow, S. Wakeman, and M. S. Jalali (2021). *Changes in characteristics of drug overdose death trends during the COVID-19 pandemic*. *International Journal of Drug Policy* **98**(Accepted), 103392.

⁶First Prize, 2020 SMDM Lee B. Lusted Prize in *Quantitative Methods & Theoretical Developments*

- J23. L. L. Czerniak, S. W. Liebel, **G.-G. P. Garcia**, M. S. Lavieri, M. A. McCrea, T. W. McAllister, and S. P. Broglio (2021). *Sensitivity and Specificity of Computer-Based Neurocognitive Tests in Sport-Related Concussion: Findings from the NCAA-DoD CARE Consortium*. *Sports Medicine* **51**(2), 351–365.
- J24. S. W. Liebel, K. Van Pelt, **G.-G. P. Garcia**, L. L. Czerniak, M. A. McCrea, T. W. McAllister, and S. P. Broglio (2020). *The Relationship between Sport-Related Concussion and Sensation-Seeking*. *International Journal of Molecular Sciences* **21**(23), 9097.
- J25. **G.-G. P. Garcia**, J. Yang, M. S. Lavieri, T. W. McAllister, M. A. McCrea, and S. P. Broglio (2020). *Optimizing Components of the Sport Concussion Assessment Tool for Acute Concussion Assessment*. *Neurosurgery* **87**(5), 971–981.
- J26. **G.-G. P. Garcia**, M. S. Lavieri, R. Jiang, M. A. McCrea, T. W. McAllister, and S. P. Broglio (2020). *Data-driven stochastic optimization approaches to determine decision thresholds for risk estimation models*. *IIEE Transactions* **52**(10), 1098–1121. ⁷
- J27. **G.-G. P. Garcia**, M. S. Lavieri, C. Andrews, X. Liu, M. P. Van Oyen, M. A. Kass, M. O. Gordon, and J. D. Stein (2019). *Accuracy of Kalman Filtering in Forecasting Visual Field and Intraocular Pressure Trajectory in Patients With Ocular Hypertension*. *JAMA Ophthalmology* **137**(12), 1416–1423.
- J28. **G.-G. P. Garcia**, M. S. Lavieri, R. Jiang, T. W. McAllister, M. A. McCrea, and S. P. Broglio (2019). *A Data-Driven Approach to Unlikely, Possible, Probable, and Definite Acute Concussion Assessment*. *Journal of Neurotrauma* **36**(10), 1571–1583.
- J29. **G.-G. P. Garcia**, K. Nitta, M. S. Lavieri, C. Andrews, X. Liu, E. Lobaza, M. P. Van Oyen, K. Sugiyama, and J. D. Stein (2019). *Using Kalman Filtering to Forecast Disease Trajectory for Patients With Normal Tension Glaucoma*. *American Journal of Ophthalmology* **199**. Publisher: Elsevier Inc., 111–119.
- J30. G. J. Schell, **G.-G. P. Garcia**, M. S. Lavieri, J. B. Sussman, and R. A. Hayward (2019). *Optimal Coinsurance Rates for a Heterogeneous Population under Inequality and Resource Constraints*. *IIEE Transactions* **51**(1), 74–91.
- J31. **G.-G. P. Garcia**, S. P. Broglio, M. S. Lavieri, M. McCrea, and T. McAllister (2018). *Quantifying the Value of Multidimensional Assessment Models for Acute Concussion: An Analysis of Data from the NCAA-DoD Care Consortium*. *Sports Medicine* **48**(7), 1739–1749. ⁸
- J32. M. H. Mofrad, **G.-G. P. Garcia**, L. M. Maillart, B. A. Norman, and J. Rajgopal (2016). *Customizing Immunization Clinic Operations to Minimize Open Vial Waste*. *Socio-Economic Planning Sciences* **54**(March 2016), 1–17.

Conference proceedings and other non-journal articles

Refereed conference papers

- C1. V. Nori-Naga, S. Sundar, and **G.-G. P. Garcia** (2023). Analysis of Machine Learning and Ensemble Methods for Forecasting COVID-19 Cases in California. In: *Proceedings of the IIEE Annual Conference & Expo 2023*, pp. New Orleans, LA
- C2. **G.-G. P. Garcia**, L. L. Czerniak, M. S. Lavieri, S. W. Liebel, M. A. McCrea, T. W. McAllister, P. F. Pasquina, and S. P. Broglio (2022). Simulation-Optimization to Distinguish Optimal Symptom Free Waiting Period for Return-to-Play Decisions in Sport-Related Concussion. In: *2022 Winter Simulation Conference (WSC)*. Singapore: IEEE, pp. 1021–1032. <https://ieeexplore.ieee.org/document/10015285/>

⁷Best Paper, 2021 IIEE Transactions Focus Issue on *Operations Engineering & Analytics*; Winner, 2020 INFORMS Minority Issues Forum Paper Competition; Winner, 2018 SMDM Lee B. Lusted Prize in *Quantitative Methods & Theoretical Developments*; First Place, 2017 INFORMS Minority Issues Forum Poster Competition

⁸Cited in American Medical Society for Sports Medicine position statement on concussion in sport (doi: 10.1136/bjsports-2018-100338)

Refereed conference abstracts

- C3. D. Chung, K. B. Adams, Y. Yang¹, **G.-G. P. Garcia**, A. Roy, X.-L. Leong, and V. Ahuja (2025). *Predicting Pediatric Surgical Case Duration Using Machine Learning: Leveraging Team Dynamics and Case Flow Attributes*. 137th Annual Meeting of The Southern Surgical Association, Hotsprings, VA
- C4. Y. Yang¹, T. Liu, C.-Y. Liao¹, S. J. Lee¹, E. Keyvanshokooh, H. Shao, M. B. Weber, F. J. Pasquel, and **G.-G. P. Garcia** (2024). *Development and Fairness Evaluation of CVD Risk Prediction Models for Patients with Type-2 Diabetes*. Society for Medical Decision Making 46th Annual Meeting, Boston, MA
- C5. C.-Y. Liao¹, E. Keyvanshokooh, F. J. Pasquel, and **G.-G. P. Garcia** (2024). *Augmenting Individualized Treatment Planning via Data-Driven Clinical Role Model Selection*. Society for Medical Decision Making 46th Annual Meeting, Boston, MA
- C6. H. S. Pandey¹, B. Lahijanian², J. D. Schmidt, R. C. Lynall, S. P. Broglio, T. W. McAllister, M. A. McCrea, P. F. Pasquina, and **G.-G. P. Garcia** (2024). *Value of Baseline Testing in Concussion Diagnosis*. ACSM Annual Meeting & World Congress, Boston, MA
- C7. C.-Y. Liao¹, E. Keyvanshokooh, and **G.-G. P. Garcia** (2023). *Data-driven Counterfactual Optimization for Personalized Clinical Decision-Making*. Society for Medical Decision Making 45th Annual Meeting, Philadelphia, PA
- C8. C.-Y. Liao¹, **G.-G. P. Garcia**, K. Paynabar, and M. S. Jalali (2023). *Improving Prediction of Opioid-involved Overdose Deaths with Multivariate Spatio-Temporal Hawkes Process*. IISE Annual Conference, New Orleans, LA
- C9. V. Nori-Naga, S. Sundar, and **G.-G. P. Garcia** (2023). *Analysis of Machine Learning and Ensemble Methods for Forecasting COVID-19 Cases in California*. IISE Annual Conference, New Orleans, LA
- C10. S. J. Lee¹ and **G.-G. P. Garcia** (2022). *A Preference-sensitive Approach to Optimal Screening Problems with Multiple Objectives: Application to Breast Cancer Screening*. Society for Medical Decision Making 44th Annual Meeting, Seattle, WA
- C11. C.-Y. Liao¹, **G.-G. P. Garcia**, K. Paynabar, and M. S. Jalali (2022). *A Novel Spatio-Temporal Modeling Framework for Improving Public Health Surveillance: An Application to Opioid-related Drug Overdose Deaths*. Society for Medical Decision Making 44th Annual Meeting, Seattle, WA
- C12. **G.-G. P. Garcia**, L. N. Steimle, W. J. Marrero, and J. B. Sussman (2022). *Design, Analysis, and Optimization of Interpretable Policies for Hypertension Treatment Planning*. Society for Medical Decision Making 44th Annual Meeting, Seattle, WA
- C13. **G.-G. P. Garcia**, N. Ghaffarzadegan, and M. S. Jalali (2022). *A Model-based Analysis of Screening Selection Rates for Post-traumatic Stress Disorder*. Society for Medical Decision Making 44th Annual Meeting, Seattle, WA
- C14. **G.-G. P. Garcia**, M. S. Lavieri, T. W. McAllister, M. A. McCrea, and S. P. Broglio (2022). *Design and Assessment of a Novel Decision-Theoretic Approach to Personalize Return-to-play from Concussion*. 6th International Consensus Conference on Concussion in Sport, Amsterdam, Netherlands
- C15. D. W. Van Pelt, **G.-G. P. Garcia**, S. R. Hinds, D. R. Howell, J. McCollum, S. J. Svoboda, and K. L. Van Pelt (2022). *Daily Concussion Symptom Presence and Severity is Highly Variable in a Healthy Population Over a 14-day Monitoring Period*. 6th International Consensus Conference on Concussion in Sport, Amsterdam, Netherlands
- C16. K. L. Van Pelt, **G.-G. P. Garcia**, S. R. Hinds, D. R. Howell, J. McCollum, S. J. Svoboda, and D. W. Van Pelt (2022). *Time of Day and Day of Week Do Not Meaningfully Impact Concussion Symptom Scores In Healthy Young Adults*. 6th International Consensus Conference on Concussion in Sport, Amsterdam, Netherlands
- C17. J. B. Caccese, **G.-G. P. Garcia**, A. P. Kontos, N. L. Port, K. Valerio, S. P. Broglio, M. A. McCrea,

- T. W. McAllister, P. F. Pasquina, and T. A. Buckley (2022). *Test-retest Reliability And Efficacy Of Individual Symptoms In Concussion Management*. ACSM Annual Meeting & World Congress, San Diego, CA
- C18. **G.-G. P. Garcia**, L. N. Steimle, W. J. Marrero, and J. B. Sussman (2022). *An Analysis of Structured Optimal Policies for Hypertension Treatment Planning*. IISE Annual Conference, Seattle, WA
- C19. **G.-G. P. Garcia**, M. S. Lavieri, T. W. McAllister, M. A. McCrea, and S. P. Broglio (2020). *The Value of Incorporating Symptom-Reporting Behavior in Optimal Treatment Decisions: An Application to Return-to-Play from Sports-Related Concussion*. Society for Medical Decision Making 42nd Annual Meeting, Chicago, IL
- C20. **G.-G. P. Garcia**, M. S. Lavieri, C. Andrews, X. Liu, M. P. Van Oyen, M. A. Kass, M. O. Gordon, and J. D. Stein (2019). *Using a Machine Learning Technique Called Kalman Filtering to Forecast Conversion from Ocular Hypertension to Primary Open Angle Glaucoma*. The Association for Research in Vision and Ophthalmology, Vancouver, BC
- C21. **G.-G. P. Garcia**, M. S. Lavieri, R. Jiang, M. A. McCrea, T. W. McAllister, and S. P. Broglio (2018). *A Novel Data-Driven Framework for Improving Risk-Based Medical Diagnosis with Application to Acute Concussion Assessment*. Society for Medical Decision Making 40th Annual Meeting, Montreal, QC
- C22. M. S. Lavieri, X. Liu, **G.-G. P. Garcia**, Z. Zhou, E. Lobaza, J. Wang, K. Sugiyama, K. Nitta, C. Andrews, M. P. Van Oyen, and J. D. Stein (2017). *Using Kalman Filtering to Personalize the Monitoring of Persons with Normal Tension Glaucoma*. The Association for Research in Vision and Ophthalmology, Baltimore, MD
- C23. **G.-G. P. Garcia**, M. S. Lavieri, and S. P. Broglio (2017). *A Markov Decision Process to Model Symptom Self-Reporting Behavior in Concussion Management*. The 3rd Multidisciplinary Conference on Reinforcement Learning and Decision Making, Ann Arbor, MI
- C24. **G.-G. P. Garcia**, M. S. Lavieri, and S. P. Broglio (2016). *Quantifying Risk Thresholds for On-field Concussion Diagnosis*. Industrial and Systems Engineering Research Conference, Anaheim, CA

Books, chapters, patents, etc.

Book chapters

- B1. A. K. McNealey, W. J. Marrero, L. N. Steimle, and **G.-G. P. Garcia** (2023). “Optimization Methods for Interpretable Treatment and Screening”. In: *Encyclopedia of Optimization*. 3rd ed. Springer, Cham. https://doi.org/10.1007/978-3-030-54621-2_866-1
- B2. S. J. Lee¹, H. S. Pandey¹, and **G.-G. P. Garcia** (2023). “Designing Interpretable Machine Learning Models using Mixed Integer Programming”. In: *Encyclopedia of Optimization*. 3rd ed. Springer, Cham

Submitted papers

- S1. Y. Yang¹, T. Liu, C.-Y. Liao¹, S. J. Lee¹, E. Keyvanshokoo, H. Shao, M. B. Weber, F. J. Pasquel, and **G.-G. P. Garcia** (2025). *Development and Evaluation of Cardiovascular Disease Risk Models for Patients with Type 2 Diabetes*. (Under review)
- S2. H. S. Pandey¹, **G.-G. P. Garcia**, S. P. Broglio, M. A. McCrea, T. W. McAllister, P. F. Pasquina, and L. B. Lempke (2025). *Comprehensive Assessment under Constrained Time (CompACT): An Operationally Constrained Machine Learning Approach*. (Under revision)⁹

⁹First Prize, 2024 INFORMS Minority Issues Forum Poster Competition awarded to Himadri Pandey

- S3. **G.-G. P. Garcia** and J. C. Martinez Mori (2024). *Who's the GOAT? Sports Rankings and Data-Driven Random Walks on the Symmetric Group*. (Under revision)
- S4. C.-Y. Liao¹, E. Keyvanshokoo, F. J. Pasquel, and **G.-G. P. Garcia** (2025). *Augmenting Individualized Treatment Planning via Data-Driven Clinical Role Model Selection*. (Under review)¹⁰
- S5. **G.-G. P. Garcia**, N. Ghaffarzagdegan, and M. S. Jalali (2024). *Optimal Screening Selection Rates for Post-traumatic Stress Disorder and The Impact of Social Stigma*. (Under revision)

MISCELLANEOUS

Outside Professional Work for Compensation (1460s)

Name of Organization	Start Date	End Date	Number of Days	Description of Activity
Synaptek, LLC	9/2021	9/2023		Technical consultant

OTHER SCHOLARLY ACTIVITY

Invited lectures and seminars

- University of Washington, Department of Industrial and Systems Engineering, *Leveraging Operations Research for Responsible AI in Medicine*, December 2024
- University of California Berkeley, Department of Industrial Engineering and Operations Research, *Leveraging Operations Research for Responsible AI in Medicine*, December 2024
- Georgia Tech, Alpha Pi Mu Lunch and Learn Seminar, *Healthcare, Optimization, Machine Learning, and Everything In Between*, April 2024
- Georgia Tech, Institute for People and Technology, *Toward Fairer Type-2 Diabetes Diagnosis and Care: A Long-term and Pipeline-level View*, March 2024
- INFORMS, Junior Faculty Interest Group Seminar, *Leveraging Operations Research for Responsible AI in Medicine: Generating Clinical Role Models for Personalized Treatment Target Selection*, February 2024
- Emory University, Georgia Center for Diabetes Translation Research, *Designing Responsible Artificial Intelligence for Diabetes Care*, September 2023
- Rochester Institute of Technology, Department of Industrial and Systems Engineering, *Interpretable Policies and the Price of Interpretability in Hypertension Treatment Planning*, October 2022
- Synaptek, *Optimizing Acute Concussion Assessment: Insights from Predictive Analytics*, May 2022
- Georgia Tech, Center for Machine Learning, *An Analysis of Structured Optimal Policies for Hypertension Treatment Planning: The Tradeoff Between Optimality and Interpretability*, March 2022
- Virginia Tech, Department of Industrial and Systems Engineering, *An Analysis of Structured Optimal Policies for Hypertension Treatment Planning: The Tradeoff Between Optimality and Interpretability*, November 2021
- Harvard Medical School, MGH Institute for Technology Assessment, *Predictive and Prescriptive Analytics for Concussion Management Decisions*, August 2020
- University of Minnesota, Department of Industrial and Systems Engineering, *Predictive and Prescriptive Analytics for Concussion Management Decisions*, January 2020
- Clemson University, Department of Industrial Engineering, *Predictive and Prescriptive Analytics for Concussion Management Decisions*, January 2020

¹⁰Finalist, 2023 SMDM Lee B. Lusted Prize in *Quantitative Methods & Theoretical Developments* awarded to Che-Yi Liao

14. Purdue University, School of Industrial Engineering and Regenstrief Center for Healthcare Engineering, *Predictive and Prescriptive Analytics for Concussion Management Decisions*, December 2019
15. Georgia Tech, H. Milton Stewart School of Industrial and Systems Engineering, *Predictive and Prescriptive Analytics for Concussion Management Decisions*, December 2019
16. University of Florida, Department of Industrial and Systems Engineering, *Predictive and Prescriptive Analytics for Concussion Management Decisions*, December 2019
17. North Carolina State University, Edward P. Fitts Department of Industrial & Systems Engineering Department, *Data-driven Decision Thresholds for Risk Estimation Models with Application to Acute Concussion Assessment*, April 2019
18. University of Michigan, Department of Industrial and Operations Engineering, *Data-driven Optimization for Concussion Management Decisions*, September 2018

Presentations given at conferences¹¹

Invited talks

1. S. J. Lee¹ and **G.-G. P. Garcia** (2024). *A Data-driven Optimization Approach to Designing Parsimonious Treatment Guidelines*. INFORMS Annual Meeting (Bonder Scholars Session), Seattle, WA
2. S. J. Lee¹ and **G.-G. P. Garcia** (2024). *Composite Clustered Multi-Task Learning for Prediction of Adverse Pregnancy Outcomes*. INFORMS Annual Meeting, Seattle, WA
3. C.-Y. Liao¹, E. Keyvanshokoo, F. J. Pasquel, and **G.-G. P. Garcia** (2024). *Augmenting Individualized Treatment Planning via Data-Driven Clinical Role Model Selection*. INFORMS Annual Meeting, Seattle, WA
4. A. K. McNealey, M. E. Meredith, **G.-G. P. Garcia**, and L. N. Steimle (2024). *Race-Neutral Calculator Still Disadvantages Black Patients*. INFORMS Annual Meeting, Seattle, WA
5. **G.-G. P. Garcia** (2023). *Leveraging Operations Research and Machine Learning for Robust, Interpretable, and Equitable Models in Healthcare*. Health Systems: The Next Generation 2023, Georgia Tech, Atlanta, GA
6. H. S. Pandey¹, **G.-G. P. Garcia**, and L. B. Lempke (2023). *Optimizing Concussion Assessment: A Mixed-Integer Optimization Approach with Time Constraints*. INFORMS Annual Meeting, Phoenix, AZ
7. S. J. Lee¹ and **G.-G. P. Garcia** (2023). *Clustered Multi-Task Learning for Prediction of Adverse Pregnancy Outcomes*. INFORMS Annual Meeting, Phoenix, AZ
8. C. X. Bugg², **G. P. Garcia**, Y. Yang¹, and K. R. Siegel (2023). *Implementing Federal Food Service Guidelines in Federal and Private Worksite Cafeterias in the United States Improves Human and Planetary Health*. INFORMS Annual Meeting, Phoenix, AZ
9. C.-Y. Liao¹, **G. P. Garcia**, and E. Keyvanshokoo (2023). *Data-Driven Counterfactual Optimization for Personalized Clinical Decision-Making*. INFORMS Annual Meeting, Phoenix, AZ
10. W. J. Marrero, **G.-G. P. Garcia**, L. N. Steimle, and J. B. Sussman (2023). *Interpretable Policies and the Price of Interpretability in Hypertension Treatment Planning*. INFORMS Annual Meeting, Phoenix, AZ
11. **G.-G. P. Garcia**, L. N. Steimle, W. J. Marrero, and J. B. Sussman (2023). *Interpretable Policy Design with Application to Hypertension Treatment Planning*. INFORMS Annual Meeting (Bonder Scholars Session), Phoenix, AZ
12. S. J. Lee¹ and **G.-G. P. Garcia** (2023). *Clustered Multi-Task Learning for Prediction of Adverse Pregnancy Outcomes*. INFORMS Healthcare, Toronto, ON
13. B. Lahijanian², **G.-G. P. Garcia**, J. D. Schmidt, and R. C. Lynall (2023). *Concussion Prediction*

¹¹Presenting author is listed first

- Using Interpretable Machine Learning Models: An Evaluation of Baseline Testing in the NCAA-DoD CARE Consortium.* INFORMS Healthcare, Toronto, ON
14. **G.-G. P. Garcia**, L. L. Czerniak, M. S. Lavieri, S. W. Liebel, K. L. Van Pelt, P. F. Pasquina, M. A. McCrea, T. W. McAllister, and S. P. Broglio (2023). *Simulation Analysis for Return-to-Play from Sports-Related Concussion.* INFORMS Healthcare, Toronto, ON
 15. **G.-G. P. Garcia**, C.-Y. Liao¹, K. Paynabar, D. Zheng, Y. Xie, and M. S. Jalali (2023). *A Locally Operating Spatio-Temporal Mutually Exciting Point Process with Dynamic Network for Improving Opioid Overdose Death Prediction.* INFORMS Healthcare, Toronto, ON
 16. C.-Y. Liao¹, **G.-G. P. Garcia**, K. Paynabar, and M. S. Jalali (2023). *A Novel Spatio-Temporal Model for Improving Public Health Surveillance: An Application to Opioid-Related Overdose Deaths.* POMS Annual Meeting, Orlando, FL
 17. C.-Y. Liao¹, **G.-G. P. Garcia**, K. Paynabar, and M. S. Jalali (2023). *Improving Opioid-involved Overdose Deaths Prediction with Multivariate Spatio-Temporal Hawkes Process.* IISE Annual Conference, New Orleans, LA
 18. S. J. Lee¹ and **G.-G. P. Garcia** (2022). *A Tolerance-based Approach to Lexicographic Multi-objective Partially Observed Markov Decision Processes.* INFORMS Annual Meeting, Indianapolis, IN
 19. C.-Y. Liao¹, **G.-G. P. Garcia**, K. Paynabar, and M. S. Jalali (2022). *Improving Prediction of Opioid-involved Overdosed Deaths with Multivariate Spatio-temporal Hawkes Process.* INFORMS Annual Meeting, Indianapolis, IN
 20. **G.-G. P. Garcia**, L. N. Steimle, W. J. Marrero, and J. B. Sussman (2022). *Design and Analysis of Interpretable Policies for Hypertension Treatment Planning.* INFORMS Annual Meeting (Bonder Scholars Session), Indianapolis, IN
 21. **G.-G. P. Garcia** (2022). *Optimal Clinical Assessment and Return to Play After Concussion.* 5th Annual NCAA-US Department of Defense Grand Alliance Concussion Conference (Virtual meeting)
 22. **G.-G. P. Garcia**, L. N. Steimle, and W. J. Marrero (2021). *An Analysis of Structured Optimal Policies for Hypertension Treatment Planning: The Tradeoff Between Optimality and Interpretability.* INFORMS Annual Meeting, Anaheim, CA
 23. **G.-G. P. Garcia**, M. S. Lavieri, T. W. McAllister, M. A. McCrea, and S. P. Broglio (2021). *Estimating the Value of Incorporating Patient Behavior in Return to Play from Concussion.* INFORMS Annual Meeting (Bonder Scholars Session), Anaheim, CA
 24. **G.-G. P. Garcia**, C. DiGennaro, E. J. Stringfellow, J. Wood, N. Poellinger, and M. S. Jalali (2021). *A Multi-state Analysis of Opioid Overdose Death Trends Since the Onset of COVID-19.* INFORMS Healthcare (Invited Session), Virtual Meeting
 25. **G.-G. P. Garcia**, L. N. Steimle, and W. J. Marrero (2021). *An Analysis of Structured Optimal Policies for Hypertension Treatment Planning.* INFORMS Healthcare (Bonder Scholars Session), Virtual Meeting
 26. **G.-G. P. Garcia**, M. S. Lavieri, T. W. McAllister, M. A. McCrea, and S. P. Broglio (2020). *The Value of Incorporating Symptom-Reporting Behavior in Return-to-play from Concussion.* INFORMS Annual Meeting (Bonder Scholars Session), National Harbor, MD
 27. **G.-G. P. Garcia**, M. S. Lavieri, and S. P. Broglio (2019). *Optimizing Treatment Decisions Under Strategically Reported Symptoms.* INFORMS Annual Meeting (Bonder Scholars Session), Seattle, WA
 28. **G.-G. P. Garcia**, M. S. Lavieri, R. Jiang, T. W. McAllister, M. A. McCrea, and S. P. Broglio (2019). *Data-driven Stochastic Optimization Approaches to Determine Decision Thresholds for Risk Estimation Models.* INFORMS Annual Meeting (Invited Session), Seattle, WA
 29. **G.-G. P. Garcia**, J. Yang, M. S. Lavieri, T. W. McAllister, M. A. McCrea, and S. P. Broglio (2019). *Doing More with Less: Applying Modern Predictive Modeling Methods to Improve Acute Concussion Assessment.* INFORMS Healthcare (Bonder Scholars Session), Boston, MA

30. **G.-G. P. Garcia**, M. S. Lavieri, T. W. McAllister, M. A. McCrea, and S. P. Broglio (2019). *Optimizing Treatment Decisions Under Objective Clinical Measures and Strategically Reported Symptoms*. INFORMS Healthcare (Invited Session), Boston, MA
31. **G.-G. P. Garcia**, M. S. Lavieri, and S. P. Broglio (2018). *Optimization Approaches for Concussion Management Decisions*. INFORMS Annual Meeting (Invited Session), Phoenix, AZ
32. **G.-G. P. Garcia**, M. S. Lavieri, and S. P. Broglio (2018). *Data-driven Optimization Models for Concussion Management Decisions*. INFORMS Annual Meeting (Bonder Scholars Session), Phoenix, AZ
33. **G.-G. P. Garcia**, M. S. Lavieri, R. Jiang, M. McCrea, T. W. McAllister, S. P. Broglio, and CARE Consortium Investigators (2017). *A Novel Framework to Determine Decision Thresholds for Medical Diagnosis*. INFORMS Annual Meeting (Invited Session), Houston, TX
34. **G.-G. P. Garcia**, M. S. Lavieri, and S. P. Broglio (2016). *An Optimization Approach to Concussion Management*. INFORMS Annual Meeting (Invited Session), Nashville, TN

Contributed talks

1. J. C. Martinez Mori and **G.-G. P. Garcia** (2024). *Who is the GOAT? Sports Rankings and Random Walks on the Symmetric Group*. INFORMS Annual Meeting, Seattle, WA
2. H. S. Pandey¹, B. Lahijanian², J. D. Schmidt, R. C. Lynall, S. P. Broglio, T. W. McAllister, M. A. McCrea, P. F. Pasquina, and **G.-G. P. Garcia** (2024). *Examining Diagnostic Utility of Baseline Testing*. TREAT Sport-related Concussion Conference, Pittsburgh, PA
3. H. S. Pandey¹, L. B. Lempke, S. P. Broglio, T. W. McAllister, M. A. McCrea, P. F. Pasquina, and **G.-G. P. Garcia** (2024). *Designing Accurate Concussion Assessments under Time Constraints*. TREAT Sport-related Concussion Conference, Pittsburgh, PA
4. A. Popoola, C. X. Bugg², **G.-G. P. Garcia**, and K. Paynabar (2024). *Reducing Homelessness Among Foster Care Youth: A Predictive Approach Using Ensemble Methods*. Third Annual Workshop on AI and Analytics for Social Good, College Park, MD
5. A. K. McNealey, M. E. Meredith, **G.-G. P. Garcia**, L. N. Steimle, K. K. Stanhope, M. H. Platner, and S. L. Boulet (2024). *Race-Blind Calculator Still Disadvantages Black Patients*. Emerging Researchers National Conference in STEM, Washington D.C.
6. C. G. Turner, A. Svirsko, **G.-G. P. Garcia**, and S. W. Liebel (2022). *Trajectory Analysis of Concussion Symptoms*. INFORMS Annual Meeting, Indianapolis, IN
7. C. G. Turner, A. Svirsko, **G.-G. P. Garcia**, and S. W. Liebel (2021). *Temporal Network Architectures of Neurocognitive and Psychological Symptoms in Collegiate Athletes with Sports-related Concussion*. INFORMS Annual Meeting, Anaheim, CA
8. **G.-G. P. Garcia**, M. S. Lavieri, R. Jiang, T. W. McAllister, M. A. McCrea, and S. P. Broglio (2019). *Data-driven Stochastic Optimization Approaches to Determine Decision Thresholds for Risk Estimation Models*. ORIE Young Investigators Workshop, Cornell University, Ithaca, NY

Professional society memberships

- Production and Operations Management Society, 2025-Present
- Institute for Operations Research and the Management Sciences (INFORMS), 2015-Present
- Society for Medical Decision Making, 2018-Present
- Institute for Industrial and Systems Engineering, 2016, 2021-Present

GRADUATE STUDENTS

Chaired Doctoral Degrees

At Georgia Tech

1. **Sun Ju Lee**, Chair

Dissertation: *Optimization for Interpretable Predictive and Prescriptive Modeling in Healthcare*

Degree Program: Operations Research

Graduated: Summer 2025

Awards:

- *ISyE George Family Fellowship*, Fall 2024
- Finalist, SMDM Lee B. Lusted Prize in *Quantitative Methods and Theoretical Developments*, 2022
- President's Fellow, 2019-2023

Current position: Assistant Professor, Freeman College of Management, Bucknell University

Current Doctoral Students

At the University of Washington

1. **Jade Li**, Co-chair with Shan Liu

Status: Started Fall 2025

2. **Panus Sawetpiyakul**, Chair

Status: Started Fall 2025

At Georgia Tech

3. **Himadri Pandey**, Chair

Degree Program: Machine Learning

Status: Passed qualifying exam, Fall 2024

Awards:

- First Prize, *INFORMS Minority Issues Forum Poster Competition*, 2024
- *Georgia HIMSS David Cowan Scholarship*, 2024
- *ISyE George Family Fellowship*, Fall 2024
- *ISyE Phillip J. and Delores A. Scott Graduate Student Health and Wellness Award*, 2024

4. **Che-Yi Liao**, Co-chair with Kamran Paynabar

Degree Program: Machine Learning

Status: Passed qualifying exam, Spring 2023; Dissertation proposal examination, Spring 2024

Awards:

- First Prize, *INFORMS Bonder Scholarship*, 2024
- *Thos and Clair Muller Research Fellowship in Health Analytics and Health Systems*, 2024
- Finalist, *INFORMS Bonder Scholarship*, 2023
- Finalist, SMDM Lee B. Lusted Prize in *Quantitative Methods and Theoretical Developments*, 2023
- First Prize, SMDM Lee B. Lusted Prize in *Quantitative Methods and Theoretical Developments*, 2022
- *ISyE George Family Fellowship*, Spring 2022, Fall 2023

RESEARCH ACTIVITIES

Funded Research

Funding Agency	Title	Your role with other PI's and co-PI's	Total Amount, Your Amount, (Subcontracts if any, University Matching if any)	Dates (start - finish)
DoD	<i>Optimizing Mild Traumatic Brain Injury Diagnostics and Healthcare Costs Using a Mathematical Framework and Clinical Decision Support Tool Across the Military</i>	co-I; L. Lempke (PI, Virginia Commonwealth University), K. Cameron (co-I, West Point), B. Dengler (co-I, Fort Worth), C. Dismuke-Greer (co-I, Veterans Affairs - Palo Alto)	\$664,500/\$2,500,000	9/2025 - 8/2029
Children's Health-care of Atlanta (CHOA)	<i>Algorithm for Counteracting the Clinical Onset of Rapid Deterioration in Pediatric Patients</i>	PI 7/2024-8/2025, Collaborator 8/2025-6/2029; K. Wang (PI, Georgia Tech 8/2025-6/2029), N. Muthu (PI, CHOA)	\$2,500,000	7/2024-6/2029
NIH AIM-AHEAD	<i>Design and Assessment of Fair Algorithms for Counterfactual Explanations to Generate Digital Role Models for Patients with Type 2 Diabetes and Hypertension</i> (Grant No. 1OT2OD032581)	PI; H. Shao (co-PI, Emory University), E. Keyvanshokoo (co-PI, Texas A&M University), M.B. Weber (co-I, Emory University), F. Pasquel (co-I, Emory University)	\$300,000/\$992,472	9/2023-9/2025
Georgia Tech	<i>Toward Fairer Diagnosis and Care of Type 2 Diabetes: A Long-Term and Pipeline-Level View</i>	Co-PI; J. Ziani (PI, Georgia Tech), J. Julien (co-I, Georgia Tech)	\$6,413/\$16,034	9/2023-4/2024
Seth Bonder Foundation	<i>Bonder Scholars Research Fund</i>	PI	\$5,000	7/2023
AHRQ	<i>R03: Design, Analysis, and Optimization of Equitable and Value-based Baseline Testing Policies for Sports-Related Concussion</i> (Grant No. R03HS29422)	PI; J. Schmidt (co-I, University of Georgia), R. Lynall (co-I, University of Georgia)	\$61,407/ \$93,011	4/2023-3/2025
Georgia Tech	COVID-19 Faculty Relief Program	PI	\$8,000	2/2023-6/2023
I3 Emory SOM/ Georgia Tech Innovation Award	<i>Designing a Data-driven and Risk-specific Approach to Maternal Health Care</i> (Grant No. UL1-TR002378)	PI; L. Steimle (PI, Georgia Tech), S. Boulet (site PI, Emory University School of Medicine), M. Platner (co-PI, Emory University)	\$34,950/\$100,000	1/2022-12/2023

DOCUMENTATION OF TEACHING EFFECTIVENESS

Courses Taught & Student Evaluations

At the University of Washington

Course	Title	Quarter	Credits	Enroll	Resp	Item 1	Item 3	Item 4	Overall
IND E 311	Stochastic Models and Decision Analysis	Winter 2026	4						

At Georgia Tech

I taught three courses at Georgia Tech. The overall score reflects the interpolated median for the question: *Concerning everything, the instructor was an effective teacher.*

Semester	Course Number	Title	Response Rate	Overall
Spring 2024	ISyE 3232	Stochastic Manufacturing & Service Systems	53/54	4.80
Spring 2024	ISyE 6320 /HS 6400	Public Impact Applications of Operations Research and Management Science	17/18	4.88
Fall 2023	ISyE 8803	Special Topics: Prediction Modeling in Medical Decision Making and Health Policy	5/5	5.00
Spring 2023	ISyE 3232	Stochastic Manufacturing & Service Systems (T14)	59/63	4.89
Spring 2023	ISyE 3232	Stochastic Manufacturing & Service Systems (T15)	69/74	4.94
Spring 2022	ISyE 3232	Stochastic Manufacturing & Service Systems	57/62	4.99
Spring 2022	ISyE 8803	Special Topics: Prediction Modeling in Medical Decision Making and Health Policy	12/15	4.89

At the University of Michigan

I taught one course at the University of Michigan as the instructor on record. The overall score reflects the median for the question: *Overall, the instructor was an effective teacher.*

Semester	Course Number	Title	Response Rate	Overall
Winter 2018	IOE 316	Introduction to Markov Processes	100/112	4.71

Other Teaching Contributions

Guest Lectures

- Spring 2023, KAP 609: Concussion Pathology & Management at the University of Delaware
- Fall 2022, ISyE 6664: Stochastic Optimization at Georgia Tech

SERVICE

Departmental Service

- Co-chair, IISE Student Chapter Faculty Advisor, 2025
- Chair, Alpha Pi Mu Student Chapter Faculty Advisor, 2025
- Member, Curriculum Transformation Committee, 2025
- Member, Undergraduate Program Scholarship and Admissions Committee, 2025

Grant Review

- National Science Foundation, SCH, 2022

Ad-hoc Journal Reviews

- Technical Journals in IE/OR:
 - European Journal of Operational Research
 - Healthcare Management Science
 - IISE Transactions
 - IISE Transactions on Health Systems
 - INFORMS Journal on Applied Analytics
 - INFORMS Journal on Computing
 - International Journal of Production Research
 - Management Science
 - Manufacturing & Service Operations Management
 - Medical Decision Making
 - Naval Research Logistics
 - Omega - The International Journal of Management Science
 - Operations Research
 - Operations Research Letters
 - Production & Operations Management
- Medical and Health Policy Journals:
 - American Journal of Public Health
 - Applied Neuropsychology: Adult
 - BMJ Open Sport & Exercise Medicine
 - BMC Medical Research Methodology
 - International Journal of Sports Medicine
 - JAMA Network Open
 - Journal of Athletic Training
 - Neurosurgery
 - Scientific Reports
 - The Clinical Neuropsychologist

Society Leadership

- Councilor, INFORMS Health Applications Society 2024
- Vice President Co-chair of Communications, INFORMS Minority Issues Forum 2024

Awards Committees

- Judge, MSOM Student Paper Competition 2025
- Judge, POMS Best Healthcare Paper Award 2025
- Judge, INFORMS Poster Competition 2023
- Judge, INFORMS Health Applications Society Student Paper Competition 2023
- Judge, INFORMS Poster Competition 2021
- Judge, INFORMS Minority Issues Forum Best Paper Award committee 2021, 2025
- Member of INFORMS Seth Bonder Scholarship for Applied Operations Research in Health Services committee 2021-2023, 2025

Conference Organizing Activities

- IISE Annual Conference & Expo, 2022
- INFORMS Annual Meeting, 2017, 2020, 2023
- INFORMS Healthcare, 2019, 2021, 2023
- ISERC, 2016

Mentoring

- INFORMS Healthcare, Coffee Break Mentoring for PhD students, 2023
- INFORMS Annual Meeting, AMAZON SCOT Mentorship Program, 2021